Hotel Booking Cancellation Prediction

A significant number of hotel bookings are called off due to cancellations or no-show.

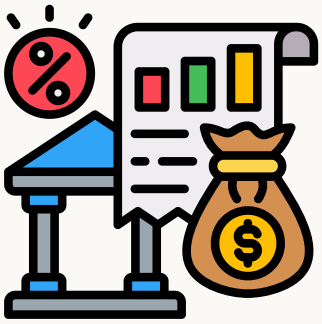
Machine Learning based solution to predict which booking is likely to be canceled.

Python | Jupyter Notebook

1. Overview of the dataset
2. Exploratory Data Analysis
3. Data Preparation for Modeling
4. Building the Model
5. Business Recommendations

BUSINESS INSIGHTS

1. Guests with higher lead times are prone to cancel.
2. A policy that establishes the penalization and refund depending on the difference between the lead time and the cancellation.
3. Implement a restriction for the lead time, not allowing a lot of time between the reservation and the arrival date,
4. When implementing a penalizing/refunding strategy, take into account the average price per room.



LESSONS & PERSPECTIVES

1. Build logistic regression, support vector machines, decision trees and random forest models.
2. Consider the guest responses to policies (dataset) in future models.

MSc. Andrea Rosas Castillo